Absolutely — below is a **full example student report** based on the *revised realistic SMR dataset* we defined earlier. This exemplar is written at a level appropriate for junior/senior nuclear engineering students and can be used as a **grading benchmark**.

You can paste this into a PDF or LMS as a reference for students.

EXEMPLAR FINAL PROJECT REPORT

NSE 351 – Introduction to Nuclear Reactor Engineering Integrated Design of a 250 MWth SMR Core
Group: A Reactor Nerds
Submitted: [Insert Date]

Executive Summary

This project presents a preliminary neutronic, thermal-hydraulic, fuel cycle, and economic analysis for a conceptual **250 MWth small modular light water reactor (LWR)** with a homogeneous cylindrical core. Core criticality was achieved using one-group and two-group diffusion theory; thermal-hydraulic limits were evaluated using simple conduction and convection models; fuel cycle performance was estimated at design burnup, and basic economic tradeoffs were considered between two enrichment options.

The core was found to be physically feasible with realistic dimensions, typical LWR flux levels, and acceptable thermal margins. Economic estimates showed favorable fuel cost per MWh, and a higher enrichment scenario offered longer cycle length at modest additional cost. Identified limitations include homogeneous core assumptions and lack of reflector or control system modeling.

---

1. Neutronic Design

1.1 One-Group Diffusion Analysis

Using homogenized macroscopic cross sections:

- $\Sigma_a = 0.018\,\text{cm}^{-1}$
- $\nu\Sigma_f = 0.021\,\text{cm}^{-1}$
- $D = 1.30 ;\text{cm} $

we compute:

$$k_\infty = \frac{0.021}{0.018} = 1.167$$

$$L=\sqrt{\frac{D}{\Sigma_a}}=\sqrt{\frac{1.30}{0.018}}=8.50\,\mathrm{cm}$$

$$B_m^2=\frac{k_\infty-1}{L^2}=\frac{0.167}{(8.50)^2}\approx 0.00231\,\mathrm{cm}^{-2}$$

With a core height ($H=350\,,\mathrm{cm}$) and extrapolated length ($H_e=350+2(0.71\,L)=362\,\mathrm{cm}$), the geometric buckling condition:

$$B_g^2=\left(\frac{2.405}{R_e}\right)^2+\left(\frac{\pi}{H_e}\right)^2$$

yields a critical extrapolated radius (R_e) of approximately **140 cm**, and physical radius ~**135 cm** after extrapolation distance. These dimensions are physically plausible for a small LWR core.

1.2 Flux Distribution

The analytic 1D radial solution indicates a flux that peaks at the center and falls towards the boundary, with normalized flux near zero at the extrapolated boundary. The shape is consistent with diffusion theory expectations. Numerical finite difference solutions agreed within ~1% of the analytic solution for a 50-node mesh.

---

2. Two-Group Diffusion

Two-group cross sections were used:

Group 1 (fast): ($D_1=1.35$, $\Sigma_{a1}=0.008$, $\nu\Sigma_{f1}=0.003$, $\Sigma_{s1\to 2}=0.020$)

Group 2 (thermal): ($D_2=0.40$, $\Sigma_{a2}=0.015$, $\nu\Sigma_{f2}=0.018$)

Solving the coupled two-group equations for a homogeneous cylinder of the dimension from Section 1.1 yields:

- Two-group ($k_{eff}\approx 1.12$)

The two-group result is lower than the one-group estimate as expected, primarily due to spectral effects and increased leakage in the fast group. The fast-to-thermal flux ratio is ~0.07, indicating a strongly thermalized core.

---

3. Thermal-Hydraulic Evaluation

3.1 Volumetric Power Density

Total reactor thermal power:

$$P=250\,\mathrm{MW}$$

Core volume:

$$V = \pi R^2 H \approx \pi (1.35\,,\mathrm{m})^2 (3.50\,\mathrm{m}) \approx 20.0\,\mathrm{m}^3$$

Volumetric heat generation:

$$q''' = \frac{250 \times 10^6}{20.0} = 12.5 \times 10^6\,\mathrm{W/m^3}$$

This is typical for LWR designs.

3.2 Fuel Centerline Temperature

For a cylindrical fuel pin with radius 0.004 m:

$$\Delta T_{max} \approx \frac{q''' R_f^2}{4 k_f} \approx \frac{12.5 \times 10^6 (0.004)^2}{4(3)} \approx 167\,\mathrm{K}$$

With inlet fuel ≈ 600 K, peak fuel temperature ≈ 767 K, well below the melting temperature of UO_2 (~2800 K), and consistent with acceptable safety margins.

3.3 Coolant Outlet Temperature

Using coolant mass flow ($\dot{m} = 1500\,\mathrm{kg/s}$) and ($c_p = 4200\,\mathrm{J/kg\text{-}K}$):

$$\Delta T_{cool} = \frac{250 \times 10^6}{\dot{m} c_p} \approx 40\,\mathrm{K}$$

Coolant outlets ~600 K, below the boiling limit (~620 K), with margin to saturation conditions.

---

4. Fuel Cycle Performance

Total heavy metal mass: 40,000 kg Target burnup: 45 MWd/kg

Total energy extractable:

$$40{,}000 \times 45 \approx 1.8 \times 10^6\,\mathrm{MWd}$$

At 250 MWth and 90% capacity factor:

$$\text{Cycle length} \approx \frac{1.8 \times 10^6}{250 \times 0.9} \approx 8\ \text{years}$$

This long theoretical cycle is a consequence of the homogeneous model without detailed control/management effects but still indicates the design provides adequate lifetime between refueling.

---

5. Economic Evaluation

Fuel component costs:

- Natural U: \$70/kg → ~\$2.8M
- Enrichment (5 SWU/kg): \$100/SWU → ~\$20M
- Fabrication: \$300/kg → ~\$12M

Fuel cost per cycle ≈ \$35M Total MWh per cycle:

$$250\,\mathrm{MW} \times 365 \times 0.9 \times 8 \approx 6.6 \times 10^{6}\,\mathrm{MWh}$$

Fuel cost per MWh ~\$5.3. Increasing enrichment to 5.5% yields ~15% longer cycle and ~8% higher fuel cost, producing slightly reduced fuel cost per MWh.

---

6. Limitations and Uncertainties

- **Homogenization:** The core was assumed homogeneous without reflector, control rods, or explicit moderator structure, which will change flux shape and criticality.
- **Leakage:** No geometric reflector reduces leakage in real designs.
- **Simplified thermal model:** No peaking factors or detailed multi-dimensional analysis.
- **Economic model:** No inclusion of capital or operating costs beyond fuel.

These simplifications should be noted in interpretation.

---

7. Conclusions and Recommendations

The conceptual 250 MWth homogeneous LWR core is physically plausible with realistic dimensions, flux levels, thermal margins, and economic figures. The design meets thermal-hydraulic criteria within conservative simplified modeling assumptions. Increasing enrichment can improve cycle length with minimal cost per MWh penalty.

Future work would include multi-group transport modeling, reflector inclusion, control rod worth evaluation, and 3D thermal-hydraulic coupling to refine safety margins.

Appendices

A – Numerical Finite Difference Code

(Included in actual submission as fully commented script.)

B – Derivations

- One-group diffusion equation
- Two-group matrix form and solution
- Thermal conduction model equations

If you'd like, I can also generate:

- A **graded solution key** mapping expected values to rubric criteria
- A **student self-assessment guide**
- A **presentation slide template** based on this exemplar

Just let me know!